

ITT320 INTRO TO COMPUTER SECURITY

Chapter 10 : Cloud and IoT Security

Universiti Teknologi MARA, Malaysia



Objectives

- What is IoT (Internet of Things)?
- How IoT Works
- IoT Ecosystem, Components
- IoT Applications and Benefits
- IoT Security & Privacy Issues
- Case Study : IoT Threats and Attacks
- IoT Security Challenges
- Cloud and IoT Security Measures
- IoT Security Tools
- IoT Security Architecture



Introduction to IoT Security

Internet of Things means every household or handy device which is used to make our world easy and better and connected with IP which transmit some data. An IoT device is simply an electronic device that is connected to the Internet.

- What are the current dangers
- Describes the most common types of attacks on your IoT Devices
- The security point of view from hackers and security professionals
- How to secure your Cloud and IoT

The Overview of IoT

The internet of things, or IoT, is a system of interrelated computing devices, mechanical and digital machines, objects, animals or people.

IoT are provided with unique identifiers (UIDs) and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction.

Hint : When devices, machines, objects, people can be conneceted to each other via the internet facilities



How Does IoT Works

1. IoT ecosystem consists of web-enabled smart devices that use embedded systems :

   a) Processors

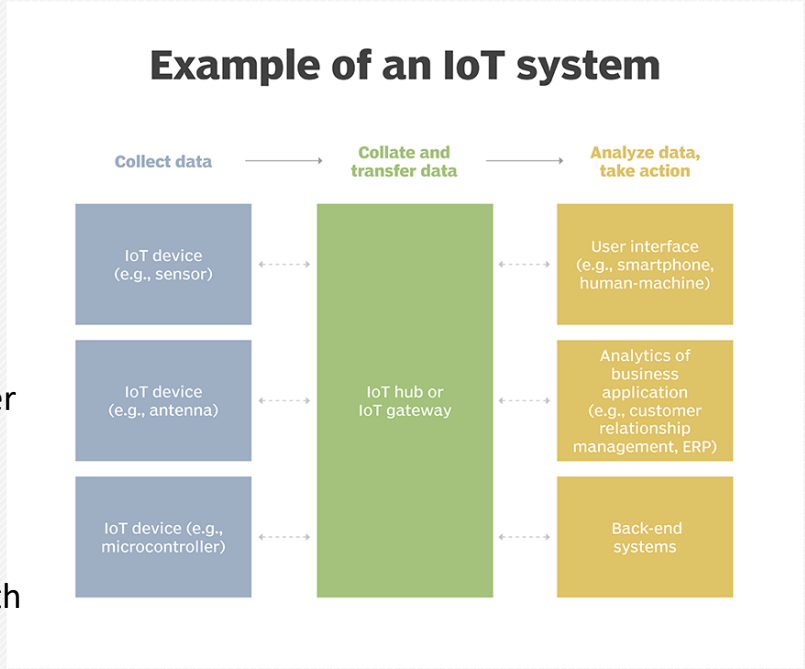
   b) Sensors

   c) Communication Hardware

   > to collect, send and act on data they acquire from their environments

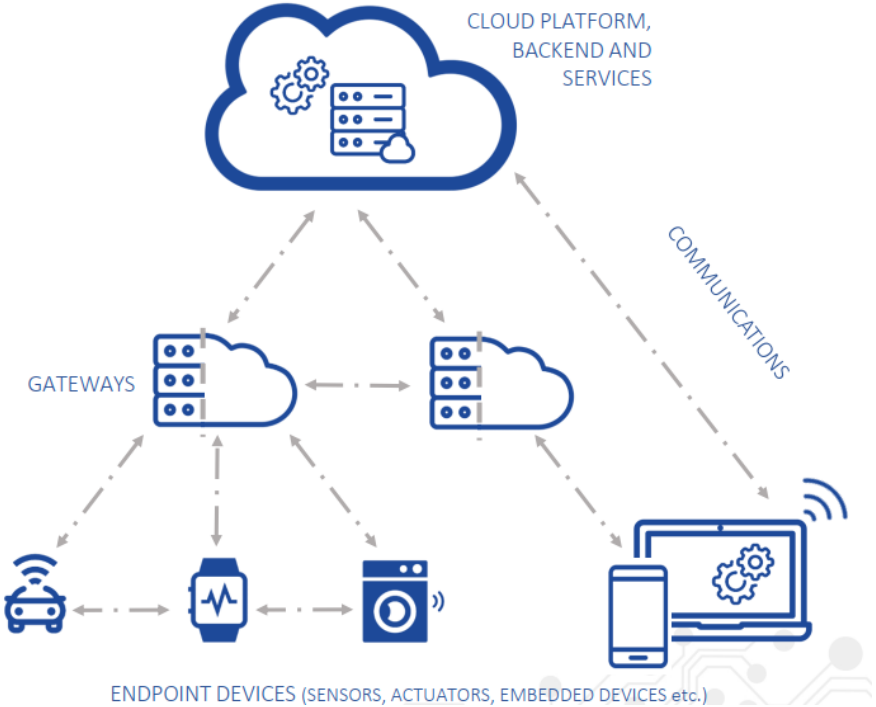
2. IoT devices share the sensor data they collect to IoT gateway or other edge device where data is either sent to the cloud to be analyzed or analyzed locally.

3. The connectivity, networking and communication protocols used with these web-enabled devices largely depend on the specific IoT applications deployed



Source : TechTarget

How Does IoT Works: IoT Ecosystem



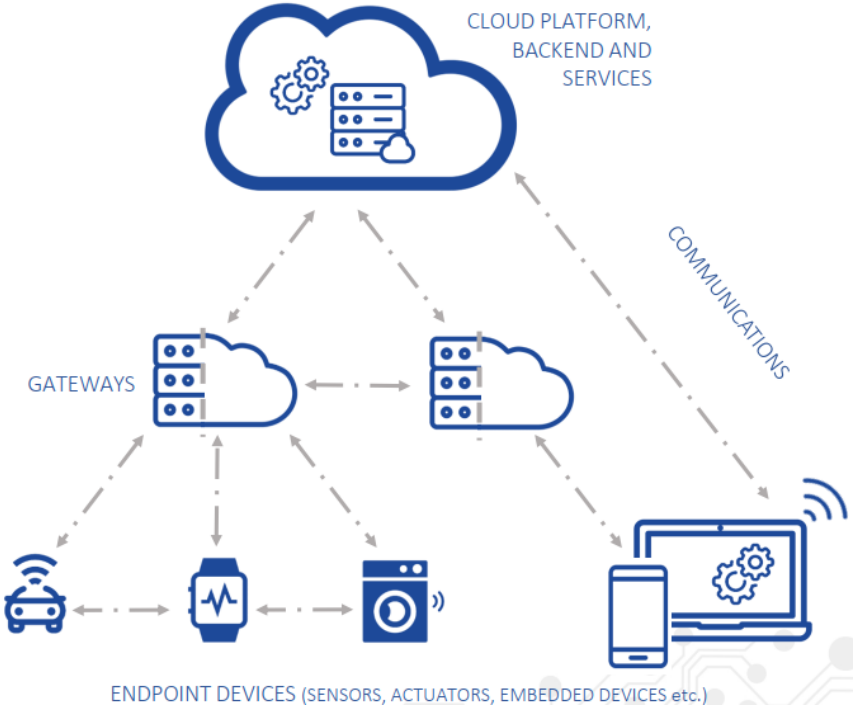
IoT Componens

1. EndPoint Devices
    a. Smart Appliances – Aircond, Vacuum Cleaner, TV
    b. Smartphone – Iphone, Android
    c. Smart 'Things' – Smart Watches, Lights, Medical Implants

2. Communications – WiFi, Zigbee, Z-Wave, NFC, RFID, BLE, LoRAWAN, MQTT/SIP/CoAP

SESSION		AMQP, CoAP, DDS, MQTT, XMPP
NETWORK	ENCAPSULATION	6LowPAN, Thread
	ROUTING	CARP, RPL
DATALINK		Bluetooth / BLE, Wi-Fi / Wi-Fi HaLow, LoRaWAN, Neul, SigFox, Z-Wave, ZigBee, USB

How Does IoT Works: IoT Ecosystem



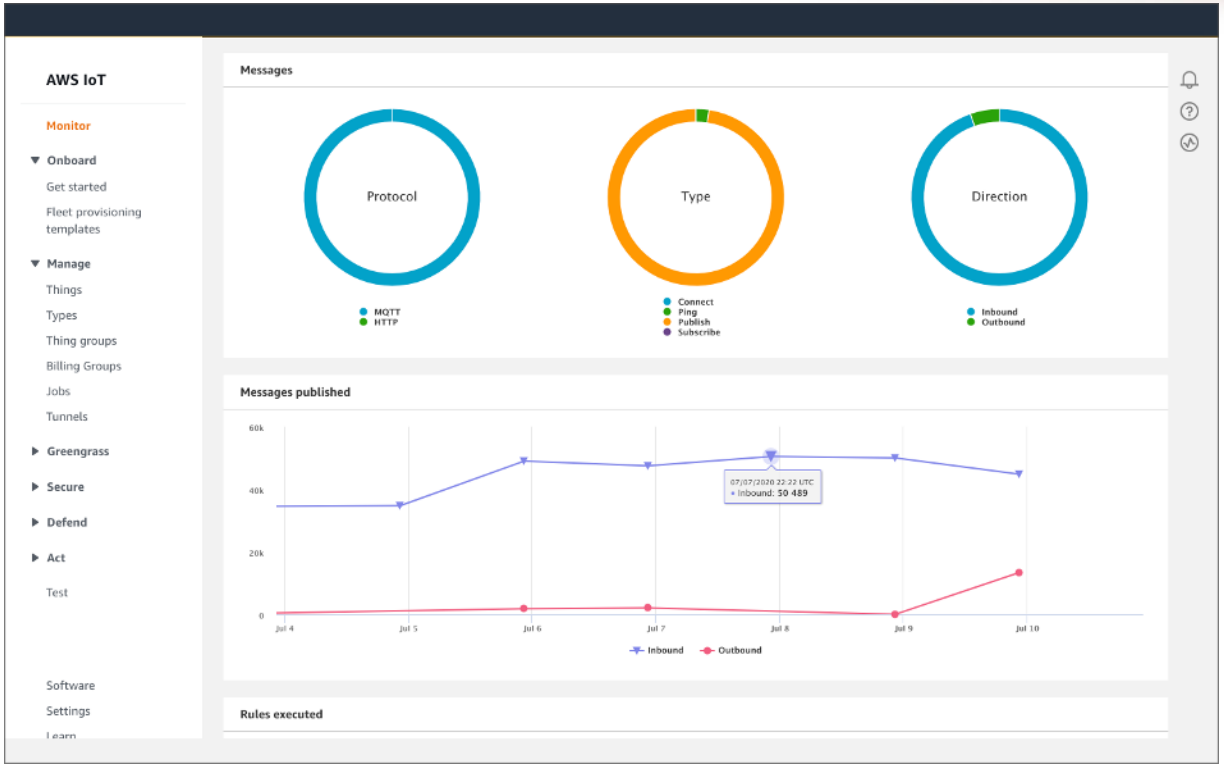
IoT Componens

3. Cloud Platform, Backend and Services

   a. Data and Storage
   b. Web-Based Services – AWS, MS Azure, AliCloud
   c. Device Management – Configuration, Settings, Maintenance, Updates.

Source : https://enisa.Europa.eu/iot

How Does IoT Works: IoT Cloud Platform



Example of IoT Cloud Platform

Amazon Web Services – AWS IoT

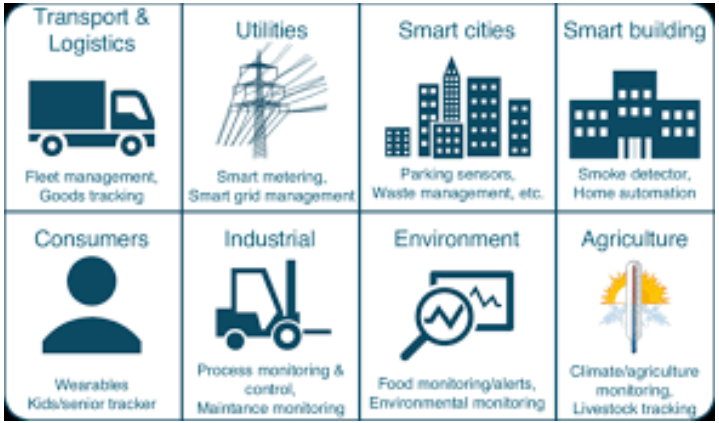
IoT
Applications & Benefits

IoT helps people live and work smarter, gain complete control over their lives.

IoT offering smart devices to automate homes, IoT is essential to business.

Some of the Benefits of IoT For People and Businesses:

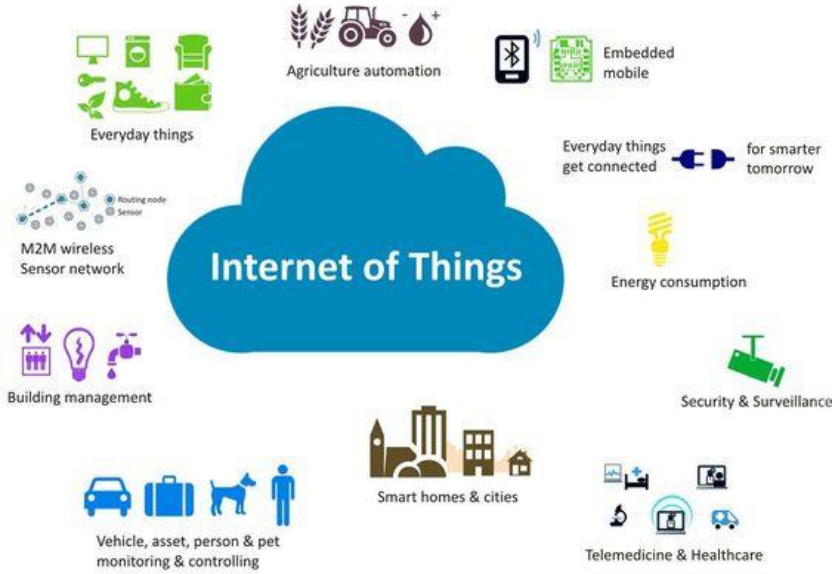
- ✓ enables companies to automate processes and reduce labor costs
- ✓ cuts down on waste and improves service delivery
- ✓ less expensive to manufacture and deliver goods
- ✓ offering transparency into customer transactions
- ✓ improve the customer experience (CX)
- ✓ Save time and money
- ✓ enhance employee productivity
- ✓ generate more revenue



IoT Applications & Benefits

There are numerous real-world applications of the internet of things, ranging from consumer IoT and enterprise IoT to manufacturing and industrial IoT. The applications of IoT:

- ✓ Consumer Segment : Smart Homes, smart thermostats, smart appliances and connected heating, lighting and electronic devices can be controlled by phone and computer
- ✓ Wearable Devices : sensors and software can collect and analyze user data
- ✓ Healthcare IoT : the ability to monitor patients more closely
- ✓ Smart Buildings : reduce energy costs using sensors , temperature can adjust automatically – Air Cond
- ✓ Agriculture : IoT-based smart farming systems can help monitor light, temperature, humidity and soil moisture
- ✓ Smart City : smart streetlights and smart meters, traffic lights

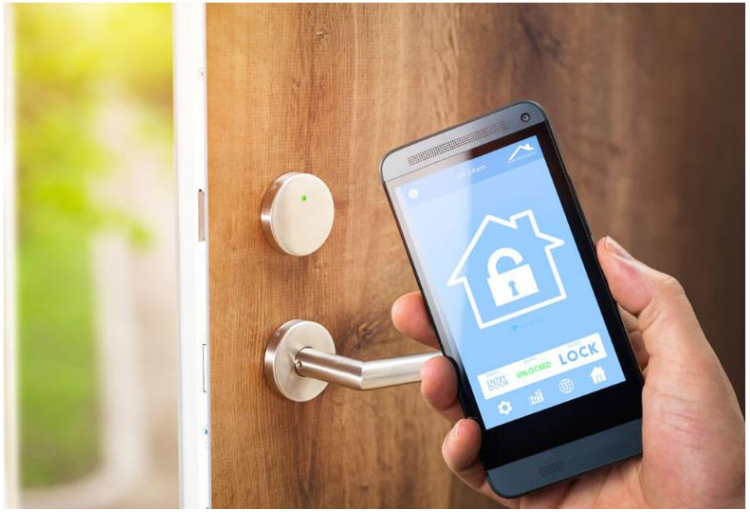


IoT Security and Privacy Issues

threatpost Podcasts / Malware / Vulnerabilities / InfoSec Insiders / Webinars

vBulletin Patches Serious Flaw in Forum Software

Bluetooth Hack Leaves Many Smart Locks, IoT Devices Vulnerable



CNBC MARKETS BUSINESS INVESTING TECH POLITICS CNBC TV INVESTING CLUB PRO

CYBERSECURITY

Hackers remotely kill Jeep's engine on highway

PUBLISHED TUE, JUL 21 2015•5:18 PM EDT | UPDATED WED, JUL 22 2015•8:41 AM EDT

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IoT Security and Privacy Issues

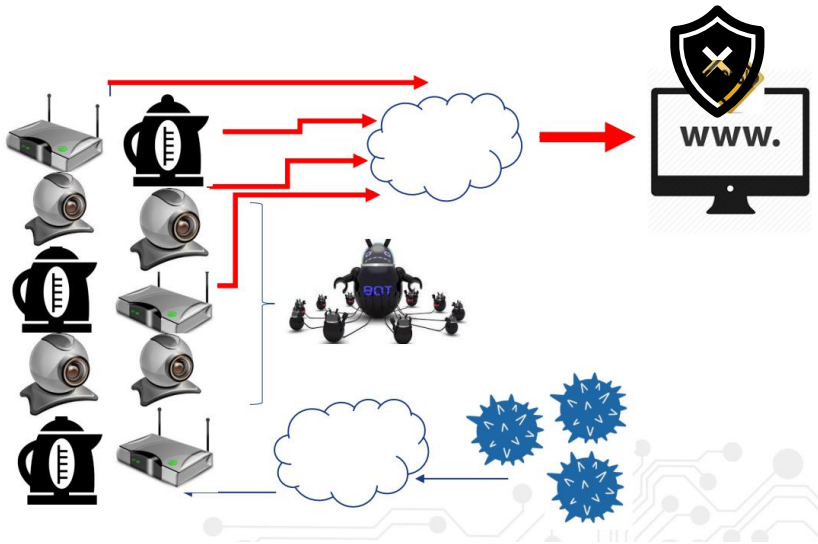




IoT Security and Privacy Issues

IoT Security – the technology area concerned with safeguarding connected devices and networks in the internet of things (**IoT**).

A number of high-profile incidents where a common IoT device was used to infiltrate and attack the larger network.



Example of IoT BotNet Attack:

- ✓ **2016, IoT attack called Mirai** – a botnet that infiltrated domain name server provider Dyn and took down many websites
- ✓ **DDoS** – an extended period of time in one of the biggest distributed denial-of-service (DDoS) attacks ever seen
- ✓ **Exploiting** - Attackers gained access to the network by exploiting poorly secured IoT devices.

- ✓ Other Attack : IoT Admin System Compromised, Command Injections

Source : https://enisa.Europa.eu/iot

Case:
Dyn Botnet DDoS Attack

- DDoS Attack in October, 2016 → Target: DNS provider **Dyn**
  - DDoS attack was staged and launched from IoT devices using the Mirai malware

- Mirai was designed for **two** main purposes:
  - Find and infect IoT devices to grow the botnet
  - Participate in DDoS attacks based on commands received by remote Command and Control (C&C) infrastructure

- Mirai operates in three stages:
  1. Infect the device
  2. Protect itself
  3. Launch attack





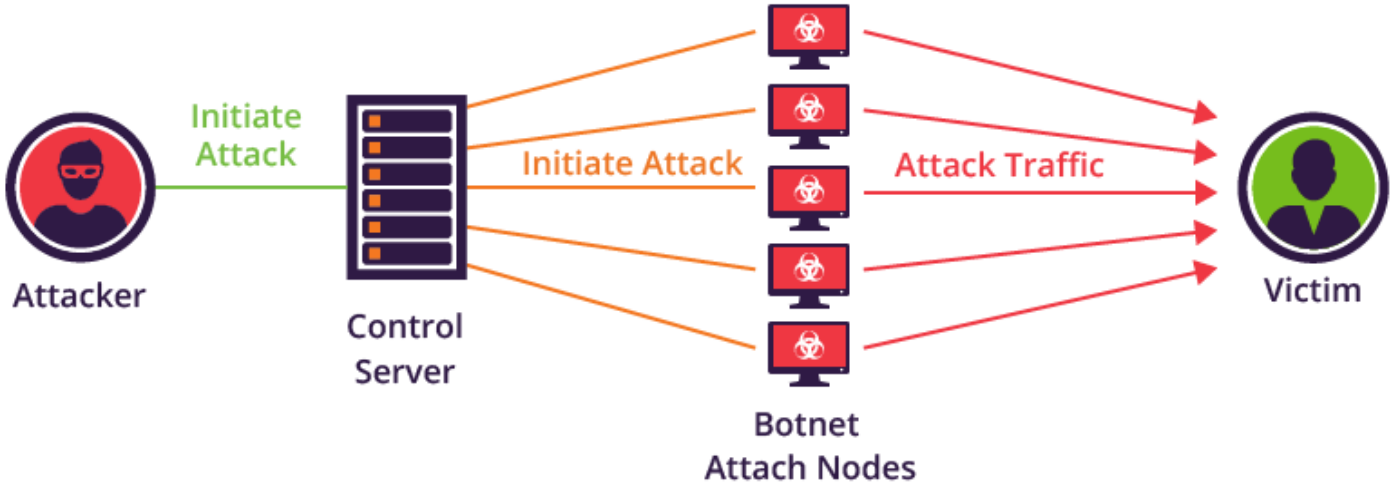
Case: Dyn Botnet DDoS Attack (Cont.)

Stage 1: Scan for IoT devices that are accessible over the Internet

- Primarily scans for ports **22, 23, 5747,** etc. that are open
- Can be configured to scan for others

- Once connected → brute-forces usernames and passwords to login to the device
- Use the device to scan networks looking for more IoT devices

Case:
Dyn Botnet DDoS Attack (Cont.)



https://www.imperva.com/blog/how-to-identify-a-mirai-style-ddos-attack/



Case: Dyn Botnet DDoS Attack (Cont.)

Stage 2: Protect itself

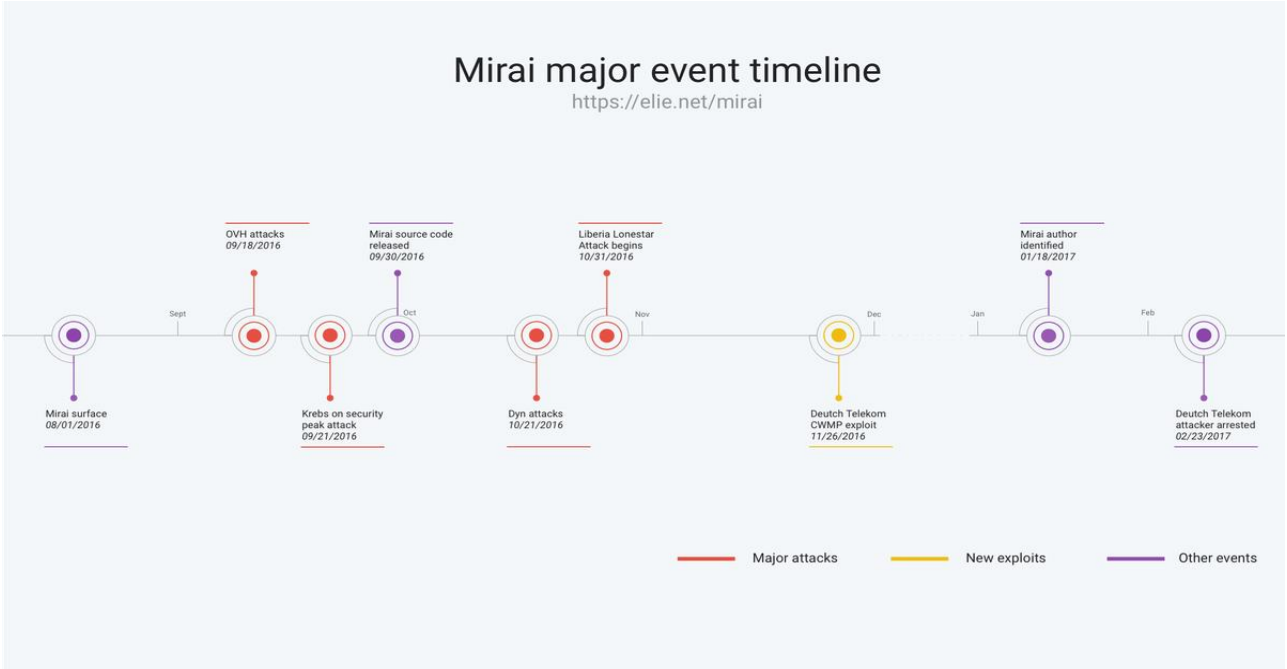
- Kill other process running on infected device (SSH, Telnet, HTTP) to prevent owner from gaining remote access to device while infected
- **Note:** Rebooting the device can remove the malware, but it can become infected again

Stage 3: Launch attack

- Infected device launches different types of attacks
- HTTP floods, SYN floods, etc. → DDoS-based attacks

- **Note: Mirai contained a list of known networks in the U.S. to avoid attacking → U.S. Postal Service, Department of Defense

Case:
Dyn Botnet DDoS Attack (Cont.)



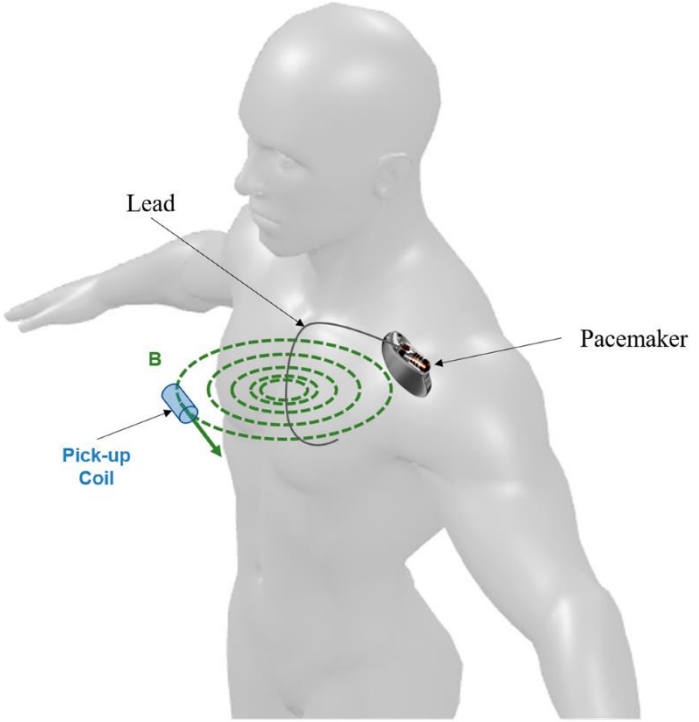
https://blog.cloudflare.com/inside-mirai-the-infamous-iot-botnet-a-retrospective-analysis/

IoT Attacks & Threats

IoT Attacks over the entire IoT Ecosystem

How and What the attacker can do with the Ecosytem, Example:

- ✓ **Sensors** – Draining the battery of pacemakers
- ✓ **Communications** – Intercepting Bluetooth LE Communications
- ✓ **Decision Making (Data Integrity)** – Modification of messages to modify smart car behavior
- ✓ **Information Privacy:** IoT Smart toys exploited to eavesdrop on children conversation



circuit breaker

This doll recorded kids' conversations without parental consent

Security experts found ways to listen in

by Ashley Carman | @ashleyrcarman | Dec 8, 2016, 11:36am EST

IoT Security - Challenges

Implementing security measures is critical to ensuring the safety of networks with IoT devices connected to them.

Main Challenges:

- ✓ Default passwords,
- ✓ Resource-constrained
- ✓ A lack of industry-accepted standards.
- ✓ Convergence of IT and operational technology (OT)
- ✓ The Rise Of Botnets
- ✓ More IoT Devices
- ✓ Lack Of Encryption
- ✓ Outdated Legacy Security



Singapore Computer Society

IoT Security – Challenges Continue

Implementing security measures is critical to ensuring the safety of networks with IoT devices connected to them.

Main Challenges:

- ✓ Weak Default Passwords
- ✓ Unreliable Threat Detection Methods
- ✓ Small Scale Attacks In IoT
- ✓ Phishing Attacks
- ✓ Inability To Predict Threats
- ✓ Infrequent Updates
- ✓ IoT Financial-Related Breaches
- ✓ User Privacy

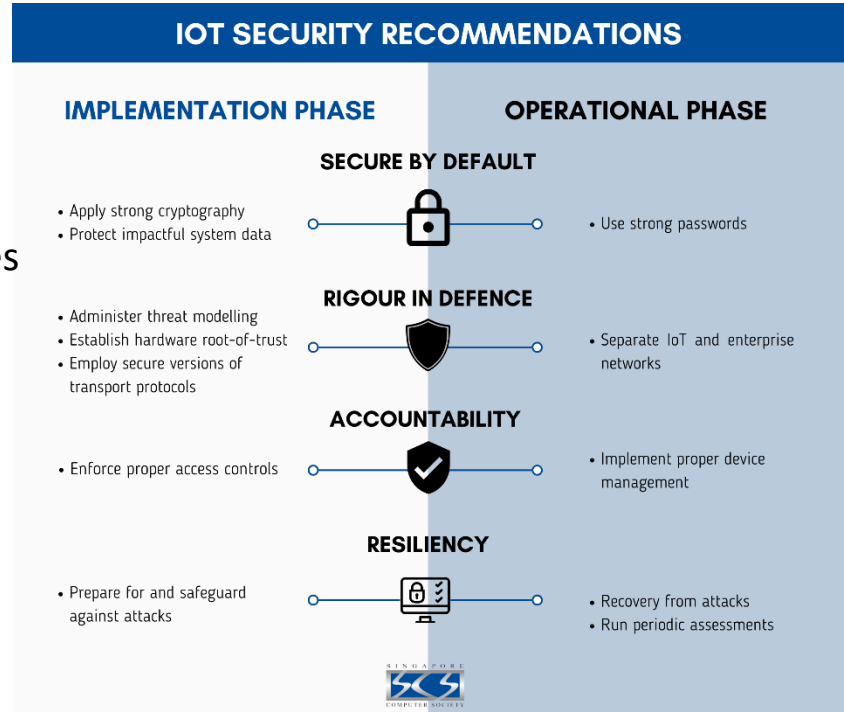


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IoT Security Measures

IoT security methods vary depending on your specific IoT application and your place in the IoT ecosystem.

- ✓ Incorporating security at the design phase
- ✓ Public key infrastructure (PKI) and 509 digital certificates
- ✓ API security
- ✓ Identity management
- ✓ Hardware security
- ✓ Strong encryption
- ✓ Network security



Source: Singapore Computer Society

IoT Security Measures Continue

IoT security methods vary depending on your specific IoT application and your place in the IoT ecosystem.

- ✓ Network access control
- ✓ Security gateways
- ✓ Patch management/continuous software updates
- ✓ Integrating teams
- ✓ Consumer education



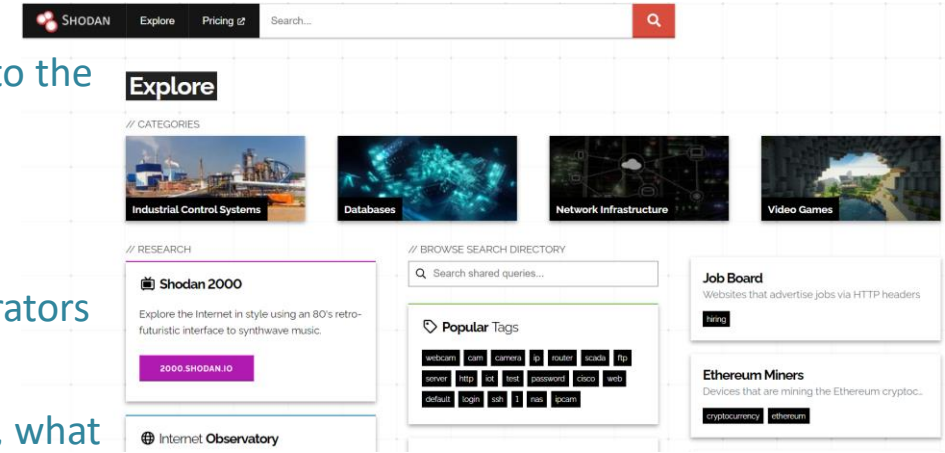
Various Computer Attack (Source: plixer.com)

IoT Attack Tools

Shodan.io

The Search Engine for IoT Devices:

1. Shodan be able to discover devices are connected to the internet, the location and who is using them
2. Monitor all computers on your network that are accessible from the internet
1. Other IoT such as power plants, Smart TVs, Refrigerators



Shodan indicate potential attacks to IoT, online scanner, what are exposing to the internet.

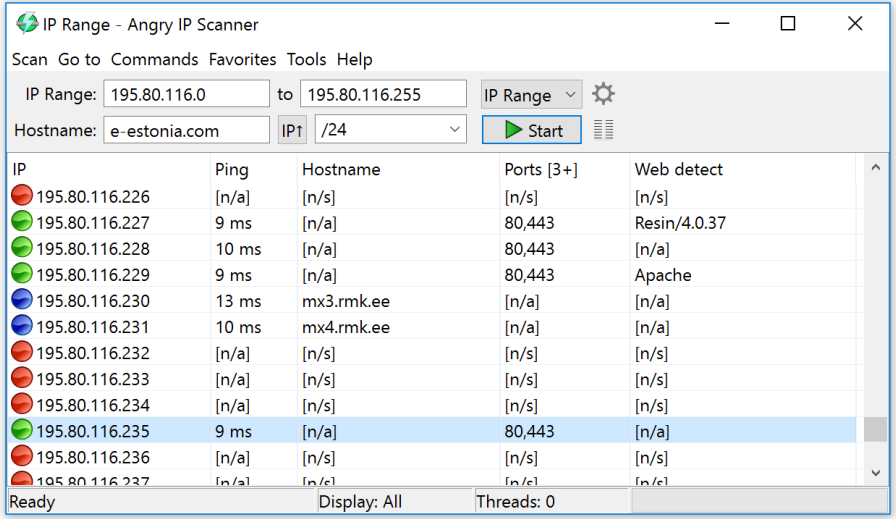
IoT Attack Tools

IP Angry

A freely available IP address and port scanner known for its ease of use, simplicity, and speed.

Angry IP scanners can scan IP addresses in any range and their ports.

1. Scans local networks as well as Internet
2. IP Range, Random or file in any format
3. Exports results into many formats
4. Extensible with many data fetchers
5. Provides command-line interface



IoT Security Architecture

- ✓ AIOTI High level Architecture functional model
- ✓ FP7-ICT IoT Architecture reference model
- ✓ NIST Network of Things
- ✓ ITU-T IoT Reference Model39
- ✓ ISO/IEC CD 30141 Internet of Things Reference Architecture
- ✓ ISACA Conceptual IoT Architecture
- ✓ oneM2M Architecture Model
- ✓ IEEE P2413 – Standard for an Architectural Framework



Vulnerability/Risk	Description	*OWASP* Top 10 IoT Risks and Vulnerabilities
1. Weak, Guessable, Hardcoded Passwords	Using easily brute-forced, publicly available, or unchangeable credentials	
2. Insecure Network Services	Unneeded or insecure network services running on the device itself, especially those exposed to the internet, compromise the C.I.A. of information or allow unauthorized remote control	
3. Insecure Ecosystem Interfaces	Insecure web, backend API, cloud, or mobile interfaces in the ecosystem outside of the device that allows compromise of the device or its related components.	
4. Lack of Secure Update Mechanism	Lack of ability to securely update the device. Examples include lack of firmware validation on device, lack of secure delivery (plaintext transmission), lack of anti-rollback mechanisms	
5. Use of Insecure or Outdated Components	Using deprecated or insecure software components/libraries that could allow the device to be compromised. Includes insecure customization of OS platforms, using third-party software, etc.	
6. Insufficient Privacy Protection	User's personal information is stored on the device and is used insecurely or without permission	
7. Insecure Data Transfer and Storage	Lack of encryption or access control of sensitive data anywhere within the ecosystem, including at rest, in transit, or during processing	
8. Lack of Device Management	Lack of security support on devices deployed within production, including asset management, update management, secure decommissioning, systems monitoring, and response capabilities	
9. Insecure Default Settings	Devices or systems shipped with insecure default settings or lack the ability to make the system more secure by restricting operators from modifying configurations	
10. Lack of Physical Hardening	Lack of physical hardening measures, allowing potential attackers to gain sensitive information that can help in future remote attacks or take local control of the device	



SUMMARY

- What is IoT (Internet of Things)?
- How IoT Works
  - Processor
  - Sensors
  - Communication Hardware
- IoT Ecosystem, Components
  - Endpoint Device
  - Communication
  - Cloud Platform
- IoT Applications and Benefits
- IoT Security & Privacy Issues



SUMMARY

- Case Study : IoT Threats and Attacks
- IoT Security Challenges
- Cloud and IoT Security Measures
- IoT Security Tools : Shodan.io , IP Angry
- IoT Security Architecture

References

https://www.techtarget.com/iotagenda/definition/Internet-of-Things-IoT

https://www.scs.org.sg/articles/iot-security-how-to-secure-your-devices

Emerging Technologies — ENISA (europa.eu)